



Quadriceps strength symmetry predicts vertical ground reaction force symmetry during running in patients who have undergone ACL reconstruction

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ABSTRACT

Objective: To determine whether quadriceps strength symmetry can predict peak vertical ground reaction force (vGRF) running force symmetry in patients who have undergone ACL reconstruction (ACLR). We also sought to determine a cutoff for quadriceps strength symmetry to identify patients at risk for vGRF running asymmetry.

Design: Retrospective cross-sectional.

Setting: Clinical facility.

Methods: Bilateral quadriceps strength and vGRF data during running were obtained from 79 patients 26–30 weeks post ACLR. Linear regression was used to determine if quadriceps strength symmetry predicted peak vGRF running force symmetry. Classification and regression tree (CART) analysis was used to determine the cutoff value for quadriceps strength symmetry to identify patients at risk for vGRF running asymmetry.

Results: Increased quadriceps strength symmetry predicted increased vGRF running symmetry ($R^2 = 0.20$). CART analysis revealed that patients with quadriceps strength symmetry less than or equal to 88% were at highest risk for vGRF running asymmetry ($R^2 = 26\%$).

Conclusion: Greater quadriceps strength symmetry is predictive of greater vGRF running force symmetry in patients who have undergone ACLR. This finding highlights the need for clinicians to consider the degree of quadriceps strength symmetry before initiating a return to running program.

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1. Introduction

Tears of the anterior cruciate ligament (ACL) are one of the most common and serious ligamentous injuries among those who participate in competitive sport (CorryWebb, 2000; Siegel et al., 2012). Although ACL reconstruction (ACLR) is recommended for patients wishing to return to their previous level of sport, the risk of sustaining a second ACL injury is 30–40 times higher compared to the risk of initial injury (Wiggins et al., 2016). In addition, persons who have undergone ACLR have been reported to exhibit early onset osteoarthritis of the tibiofemoral and patellofemoral joints. The reported prevalence of symptomatic knee osteoarthritis following ACL injury at the patellofemoral and tibiofemoral joints is

15% and 35%, respectively (Lie et al., 2019).

Although there is a 6.8–7.7 fold increase in the odds for the development of knee osteoarthritis following ACLR (Webster & Hewett, 2022), underlying mechanisms remain unclear. However, altered running mechanics following ACLR have been hypothesized to contribute abnormal joint loading and long-term joint degeneration (Tashman et al., 2004). Abnormal running mechanics following ACLR have been reported 3 months to 5 years post-surgery, with the most commonly observed asymmetries occurring in the sagittal plane (knee flexion angles and knee extensor moments) (Pairot-de-Fontenay et al., 2019). A global indicator of altered lower extremity loading is the vertical ground reaction force (vGRF) asymmetry which has been shown to be more pronounced in patients <9 months post ACLR compared to > 9 months post ACLR (Thomson et al., 2018).

The quadriceps muscle group plays an important role during the deceleration phase of running. Specifically, the knee extensors function eccentrically to control knee flexion and account for 41% of

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the negative lower limb work during typical rearfoot strike running (Stearne et al., 2014). Not surprisingly, previous studies have reported that altered quadriceps function (i.e., diminished strength or rate of torque development) adversely affects running kinematics and kinetics (Kline et al., 2015; Lewek et al., 2002; Pamukoff et al., 2018).

The decision regarding return to running is made early in the rehabilitation process (median time = 12 weeks) (Rambaud et al., 2018). Based on a recent systematic review on criteria for return to running following ACLR, there is no standard assessment for this key decision (Rambaud et al., 2018). Although quadriceps strength symmetry thresholds to signal clearance for return to running range from 60% to 85% (with 80% being the most common), research supporting specific strength requirements is limited (Rambaud et al., 2018).

To date, only one study has attempted to provide evidence for specific quadriceps strength symmetry requirements for return to running. Read et al. dichotomized patients based on 4 different quadriceps strength symmetry cutoffs (65%, 70%, 75%, 80%) and found no differences in peak vGRF running symmetry between strength groups. As such, these authors concluded that there remains insufficient evidence to prescribe a specific ‘minimum threshold’ for quadriceps strength symmetry when the goal is to improve running mechanics (Read et al., 2021). However, a limitation of this study was the fact that patients were examined relatively early in their recovery (average = 19.1 weeks) and running speed was standardized across all patients (10 km/h). It is possible that vGRF asymmetry during the time frame at which running is first initiated may be influenced by psychological factors such as fear avoidance and confidence. In addition, the relatively slow running speed employed by Read et al. may have resulted in an abnormal stride pattern for some participants thus limiting the ability to identify a relationship between quadriceps strength symmetry and running force symmetry.

To date, it remains unclear whether quadriceps strength asymmetry underlies vGRF asymmetry during running. In addition, there is insufficient evidence to prescribe a specific ‘minimum threshold’ for quadriceps strength symmetry, when the goal is to minimize vGRF asymmetry. The primary purpose of this study was to determine whether quadriceps strength symmetry can predict peak vGRF running symmetry in patients who have undergone ACLR. In addition, we sought to determine a cutoff for quadriceps strength symmetry that may protect against high peak vGRF asymmetry during running. Information gained from this study will aid clinicians in making more informed decisions about what is an appropriate quadriceps strength symmetry threshold to return to running post-ACLR.

2. Methods

2.1. Patients

We performed a retrospective analysis of patients who completed strength testing and a running evaluation at a single outpatient clinical facility. A sample size calculation using G*Power for linear regression using a 5% significance level, 90% power, 2 possible predictors, and a medium effect size ($R^2 = 0.20$), revealed that 54 participants were required (Faul et al., 2007).

Patients who were assessed 26–30 weeks post ACLR (28.1 ± 1.3 weeks) and were capable of a self-selected running speed greater than 5 mph were included in this retrospective analysis ($N = 79$). This timeframe was selected as it was assumed that patients would be more comfortable loading their limb at this point in their rehabilitation, thus avoiding potential confounding impact of psychological factors (i.e., confidence, fear avoidance) on running

symmetry. Patients who met the above criteria were included in the study regardless of age, mechanism of injury, graft type or presence of any co-existing pathologies (i.e., meniscal tears, etc.).

Prior to injury, all eligible patients had been involved in sports that required running related activities. Descriptive statistics for the patient sample are presented in Table 1. ---The University of Southern California Institutional Review Board of the Health Science Campus approved this retrospective analysis.

2.2. Procedures

Patients underwent isometric quadriceps strength testing and vertical ground reaction force (vGRF) symmetry analysis during running 26–30 weeks post-ACLR as part of normal clinical care. Rehabilitation was not standardized as patients were treated at various clinical facilities. Prior to the data collection procedures described below, patients warmed up on a bicycle for approximately 5 min.

2.2.1. Quadriceps strength assessment

A motor-driven dynamometer (BTE Primus; BTE, Baltimore, MD, USA) was used to assess isometric quadriceps strength bilaterally. Patients were seated on a chair and straps were used to secure the thigh of the testing limb. The rotational axis of the unit was aligned with the lateral femoral condyle, and the resistance pad of the lever arm was positioned above the malleoli. Bilateral quadriceps strength was assessed isometrically at 60 degrees of knee flexion. Sixty degrees of knee flexion was used as this position has been found to result in the greatest torque output in healthy females (Knapik et al., 1983). Isometric quadriceps strength was evaluated as this measure of muscle performance previously has been used as part of return to running decisions post-ACLR (Rambaud et al., 2018).

Three maximal contractions (approximately 5 s in duration) were performed for each limb with verbal encouragement. Thirty seconds of rest was provided between trials. The peak torque value from each of the 3 trials was averaged for each limb. Quadriceps strength symmetry was calculated by dividing the average peak torque of the involved limb by the average peak torque of the uninvolved limb.

2.2.2. Running assessment

Vertical ground reaction forces during running were collected using the Zebris FDM-THQ-S-3i Treadmill (Zebris Medial GmbH, Germany). The system consists of a treadmill ergometer (H/P/Cosmos, Quasar med, H-P-Cosmos Sports & Medical GmbH, Nussdorf-Traunstein, Germany) with an integrated, calibrated measuring sensor matrix (pressure platform) embedded beneath the belt. The pressure platform contained 10,240 sensors with a 135.5×54.1 cm sensor area.

The peak vGRF was derived from the pressure data within the

Table 1
Participant characteristics (N = 79).

Age (yrs)	22.9 ± 11.5 (12–58)
Mass (kg)	66.5 ± 12.0 (45.5–107.3)
Height (m)	1.70 ± 0.10 (1.50–1.93)
Female	55 (69.6%)
Self-Selected Running Speed (mph)	7.3 ± 1.3 (5.5–10.5)
Time post ACLR (weeks)	28.1 ± 1.3 (25.9–30.3)
Graft Type	
Allograft	26 (32.9%)
Autograft	53 (67.1%)

Values are presented as Mean ± SD (Range) for continuous variables and number (percentage) for categorical variables.

Zebris proprietary software as described by others (Hollander et al., 2020; Reed et al., 2013; Van Alsenoy et al., 2019). The vGRF was estimated from the pressure data by multiplying each pressure signal by its cross-sectional area, and then summing across all pressure sensors. Data were collected at a sampling rate of 120 Hz. The Zebris treadmill system has been shown to be reliable and valid for measuring kinetic and temporal variables during walking and running (Reed et al., 2013; Van Alsenoy et al., 2019).

Running speed for each patient was self-selected. Treadmill speed was gradually increased over the course of 1–2 min until patients felt comfortable with the speed based on self-report. After 2 min of running at their self-selected speed, data were captured for 30 s. Running force symmetry was calculated by dividing the peak vertical ground reaction force (vGRF) during stance on the involved limb (N) by the peak vGRF during stance on the uninvolved limb (N). The force symmetry calculation was performed for each foot contact during the 30 s data collection period and averaged for statistical analysis. Sample vGRF curves are shown in Fig. 1.

2.2.3. Statistical analysis

Paired t-tests were used to compare limb differences in quadriceps strength and the vGRF during running. Univariate linear regression analysis was used to determine if quadriceps strength symmetry (independent variable) predicted vGRF running symmetry (dependent variable). The model was evaluated for potential confounders (running speed, sex, age, or graft type). Any variable that altered the beta coefficient for quadriceps strength symmetry by a minimum of 10% was deemed a confounder and added to the model.

To determine the quadriceps strength symmetry cutoff that differentiated between patients with high vs. low vGRF running symmetry, classification and regression tree (CART) analysis was used. Specifically, a decision tree with a single level was created to predict vGRF running symmetry from quadriceps strength symmetry based on the cutoff defined by the algorithm. The R^2 for the CART model was calculated by regressing the predicted probabilities

against vGRF running symmetry. Statistical significance was defined as $p < 0.05$. All analyses were performed using SPSS statistical software (SPSS Inc., Chicago, Ill; Version 27) and a custom Matlab script (The Mathworks, Inc., Natick, MA) with alpha set at 0.05.

3. Results

Descriptive statistics for quadriceps strength and the peak vGRF during running (unilateral and symmetry) are presented in Table 2. On average, quadriceps strength values in the involved limb were lower than the non-involved limb (2.15 vs. 2.47 Nm/kg, $p < 0.001$). In addition, peak vGRF values in the involved limb were lower than the non-involved limb (21.30 vs. 22.17 N/kg, $p < 0.001$).

Increased quadriceps strength symmetry was found to significantly predict increased vGRF running symmetry ($p < 0.001$, $R^2 = 0.20$) (Fig. 2). No confounders were identified. CART analysis identified the quadriceps strength symmetry cutoff to differentiate patients with high and low vGRF symmetry was 0.88 (Fig. 3). Specifically, those with quadriceps strength symmetry ≤ 0.88 were predicted as having low vGRF symmetry while those with quadriceps strength > 0.88 were predicted as having high vGRF symmetry. The R^2 for the CART model was 26%.

4. Discussion

Running asymmetry following ACL reconstruction (ACLR) has been reported to persist up to 5 years post-surgery (Pairot-de-Fontenay et al., 2019). This is concerning as abnormal running mechanics have been hypothesized to contribute long-term joint degeneration (Tashman et al., 2004). Although quadriceps strength asymmetry has been proposed as being contributory to running asymmetry in this population (Pairot-de-Fontenay et al., 2019), there remains insufficient evidence for a ‘minimum threshold’ for quadriceps strength symmetry to minimize running asymmetry following ACLR (Rambaud et al., 2018; Read et al., 2021). As such, the primary purpose of this study was to determine whether quadriceps strength symmetry can predict peak vGRF running force symmetry in persons who have undergone ACLR. In addition, we sought to determine a cutoff for quadriceps strength symmetry to identify patients at risk for vGRF running asymmetry.

In terms of the primary aim, our results indicated that greater quadriceps strength symmetry predicted greater running force vGRF symmetry in patients 26–30 weeks post ACLR. However, only 20% of the variance in the running force vGRF symmetry was explained by quadriceps strength symmetry. Other factors not evaluated in this study, such as footfall pattern, hip strength, psychological readiness, and kinematic/kinetics, may be sources of unexplained variance.

Our findings are in contrast to Read et al. who reported no association between quadriceps strength symmetry and vGRF symmetry in persons post ACLR (Read et al., 2021). As noted above, Read et al. evaluated patients early in the rehabilitation process (average time = 19.1 weeks) at a fixed running speed (Read et al., 2021). In the current study, we evaluated patients at a later time period to avoid the potential confounding influence of fear and confidence on limb loading. This is an important consideration given that psychological (i.e., confidence to return to play) and physical factors (i.e., quadriceps strength symmetry) have been shown to be positively correlated in patients following ACLR within the first 12 months post-surgery (Della Villa et al., 2021; Straub et al., 2021). In addition, psychological readiness to return to play increases over time (Langford et al., 2009), suggesting that psychological factors may be less impactful on running symmetry later in the rehabilitation process.

In terms of the secondary aim of our study, it was determined

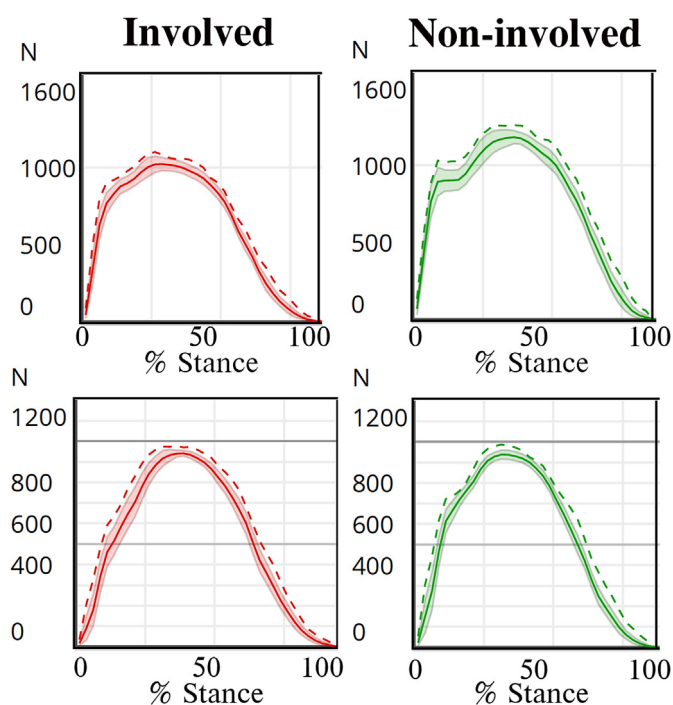


Fig. 1. Sample vGRF curves for a patient with high vGRF running asymmetry (0.86, top row) vs. low vGRF running asymmetry (1.0, bottom row). Dotted line represents 1 standard deviation from the mean (solid line).

Table 2
Quadriceps Strength and vGRF Running Data.

Quadriceps Strength	
Involved limb (Nm/kg)	2.15 ± 0.47 (1.23–3.20)
Non-involved limb (Nm/kg)	2.47 ± 0.44 (1.55–3.45)
Symmetry	0.87 ± 0.12 (0.56–1.29)
Peak vGRF Running	
Involved limb (N/kg)	21.30 ± 3.91 (14.96–33.12)
Non-involved limb (N/kg)	22.17 ± 3.84 (16.47–33.52)
Symmetry	0.96 ± 0.06 (0.86–1.11)

Values are presented as Mean ± SD (Range). Abbreviations, vGRF: vertical ground reaction force.

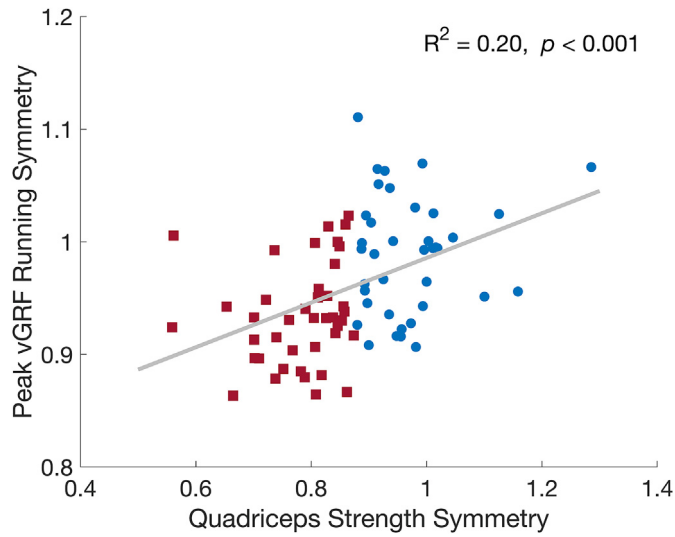


Fig. 2. Quadriceps strength symmetry predicts peak vGRF running symmetry 26–30 weeks post ACLR. Patients with quadriceps strength symmetry ≤ 0.88 predicted as having low vGRF running symmetry (red squares). Patients with quadriceps strength symmetry >0.88 predicted as having high vGRF running symmetry (blue circles). Abbreviations, vGRF: vertical ground reaction force; ACLR: ACL reconstruction. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

that a quadriceps strength symmetry value of 88% could differentiate between patients with low vs. high peak vGRF asymmetry. On average, patients with quadriceps strength symmetry $\leq 88\%$ had a 7% deficit in running symmetry compared to patients with quadriceps strength symmetry $>88\%$ who had a 1% deficit. The CART model explained 26% of the variance in vGRF running symmetry, which was slightly higher than that observed from linear regression.

Our finding of quadriceps strength asymmetry being contributory to vGRF asymmetry during running is consistent, in part, with previous studies that have linked quadriceps strength asymmetry to vGRF asymmetry during landing tasks. Patients with low quadriceps symmetry ($<85\%$) have been found to display greater asymmetry in vGRF variables (peak force and loading rate) during a double-leg drop jump compared to those with high quadriceps symmetry ($\geq 90\%$) and healthy individuals (Schmitt et al., 2015). However, during a single-leg drop land, patients with low quadriceps strength symmetry ($<85\%$) have not been reported to exhibit greater asymmetry in the aforementioned vGRF variables (Ithurburn et al., 2015). It was postulated that patients with low quadriceps strength symmetry may compensate with increased trunk flexion to decrease knee extensor demands on the involved limb (Ithurburn et al., 2015). Given that running also is considered a single limb impact activity, it is possible that some patients in the current study displayed compensatory behaviors to achieve vGRF

Peak vGRF Running Symmetry

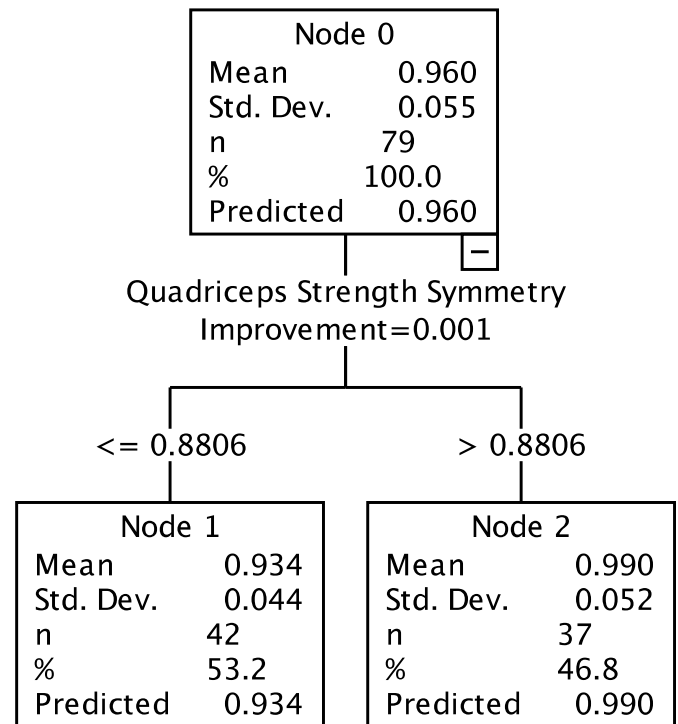


Fig. 3. Decision tree derived using CART analysis to calculate cutoff for quadriceps strength symmetry to differentiate between high vs. low peak vGRF running symmetry. vGRF: vertical ground reaction force; CART: classification and regression tree.

symmetry despite having quadriceps strength asymmetry. The premise is supported by the fact that the average vGRF symmetry in the current study was considerably higher than the average quadriceps symmetry (96% vs 87%, respectively).

The findings of the current study highlight the need for clinicians to consider the degree of quadriceps strength symmetry before initiating a return to running program. This recommendation is consistent with previous research that has shown that quadriceps strength asymmetry is linked to poorer hop performance (Schmitt et al., 2012), underloading of the involved limb during landing (Ithurburn et al., 2015), asymmetry in peak vGRF during landing (Schmitt et al., 2015), and altered gait (Lewek et al., 2002). In addition, our results indicate that a quadriceps strength symmetry value $\leq 88\%$ may be suggestive of vGRF asymmetry during running. It should be noted however that this value is somewhat higher than what has been reported as possible strength thresholds to signal return to running (60%–85%, with 80% being the most common) (Rambaud et al., 2018). Interestingly, the 88% quadriceps strength symmetry cutoff identified in the current study is consistent with data suggesting that $>90\%$ symmetry in quadriceps strength reduces the risk for a second ACL injury (Grindem et al., 2016).

As noted above, many ACL protocols initiate running/jogging at 3–4 months (Rambaud et al., 2018), a time-period in which many patients have not yet achieved quadriceps symmetry. This potentially is problematic as loading asymmetry may become learned as evidenced by the fact that vGRF landing asymmetry has been found to persist 2 years or more after ACLR (Paterno et al., 2007). An argument could be made that returning patients to running prematurely may have unintended long-term consequences.

An important limitation of using quadriceps strength symmetry

as a primary outcome variable is the fact that weakness in the non-involved limb could mask involved limb weakness. Non-involved limb quadriceps strength deficits have been reported in persons who have undergone ACLR, compared with healthy controls, up to 24 months after ACLR (Chung et al., 2015). Although the current study did not evaluate unilateral strength thresholds related to running symmetry, future research should establish whether unilateral strength measures can be used to infer altered running mechanics post-ACLR. Nonetheless, the results presented here do have clinical value as the use of quadriceps strength symmetry remains a primary standard for most return to sport criteria (including return to running) (Kaplan & Witvrouw, 2019; Rambaud et al., 2018).

The current study has several limitations that should be considered. First, our determination of running asymmetry was based on only a single variable (peak vGRF). As such, the quadriceps strength cutoff for distinguishing between patients with low vs. high running vGRF may not be consistent among related running variables that can be used to assess asymmetry (e.g., knee flexion angle and/or knee extensor moment). Second, quadriceps strength symmetry was evaluated isometrically. More robust relationships may have been evident if other measures of muscle performance were evaluated (i.e., eccentric strength). Third, although we included confounded checks for numerous variables (running speed, gender, age, or graft type), rehabilitation was not standardized as patients were treated at various clinical facilities. Lastly, our results are applicable to only patients 26–30 weeks post ACLR. Therefore, caution is advised when trying to extrapolate our results to patients evaluated at time points outside of this timeframe.

5. Conclusion

Greater quadriceps strength symmetry predicted greater running force symmetry in patients who have undergone ACLR (26–30 weeks). Patients who have quadriceps strength symmetry less than or equal to 88% may be at risk for vGRF running asymmetry. This strength threshold should be considered when clearing patient to return to running post ACLR.

Ethical statement

Work for this scientific research study with human subjects was approved by the Institutional Review Board of the Health Sciences Campus at the University of Southern California (#HS-17-00253) on 11/3/2020.

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Declaration of competing interest

None.

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